

List of Sensor Data

Hydrodynamic Data

#	Parameter	Sensor	Unit	Description
1	Depth	Echo Sounder	Meters	Distance from the surface/hull to the seabed
2	SOG	Speed log	Knots	Actual speed over the earth's surface.
3	Wind speed	anemometer	knots	How fast the water is moving, and where it is flowing
4	Wind Direction	Wind vane	degree	
5	Wave Period	IMU or MRU	Seconds	The time (in seconds) between two wave crests
6	Water current speed	ADCP	Knots	How fast the wind is blowing, and where it is coming <i>from</i>
7	Water current direction	ADCP	degree	
8	Latitude	GPS	Degree	The ship's exact GPS coordinates on the globe
9	Longitude	GPS	Degree	
10	Heading	Gyrocompass	Degree	Direction to the front of the ship is pointing out.
11	Water Density	Hydrometer	Kg/m ³	How heavy the water is (affects how deep the ship floats)
12	Pitch Angle	IMU	degree	pitch angle is the tilt along the bow-to-stern (front-to-back) axis
13	COG	GPS/GNNS		

Seakeeping Input Parameters

IMU Sensor

ROS message: sensor_msgs/msg/Imu | **Published frequency:** 50 Hz | **Internal processing frequency:** 10 Hz

#	Parameter	Description	Unit	Source / Calculation
1	True Roll	Ship's side-to-side tilt	degrees	Derived from IMU orientation quaternion
2	True Pitch	Ship's front-to-back tilt	degrees	Derived from IMU orientation quaternion
3	True Yaw	Ship's heading deviation	degrees	Derived from IMU orientation quaternion

Navigation Sensor

ROS message: nav_msgs/msg/Odometry | **Published frequency:** 50 Hz | **Internal processing frequency:** 10 Hz

#	Parameter	Description	Unit	Source / Calculation
4	Surge Velocity	Forward speed variation	m/s	twist.linear.x
5	Sway Velocity	Lateral drift velocity	m/s	twist.linear.y
6	Speed	Current speed over ground	knots	Magnitude of velocity vector converted to knots
7	Heading	Current compass heading	degrees (0°–360°)	Yaw from pose.orientation quaternion
8	Speed (STW)	Speed Through Water; primary preferred speed input over SOG	knots	Speed relative to the water column
9	Rudder Angle	Ship's command or actual rudder	degrees	Steering gear / Autopilot sensor

		deflection angle		
10	Engine RPM	Raw engine rotational speed	RPM	Propulsion control system / Engine tachometer

Speed input preference: Speed Through Water (STW) is the primary preferred speed input because it represents vessel motion relative to the surrounding water. If STW is not available, use Speed or Speed Over Ground (SOG) as the fallback input.

Wave Sensor

ROS message: Custom topic or std_msgs/msg/String | **Published frequency:** 10 Hz

#	Parameter	Description	Unit
9	Wave Elevation at Bow	Live wave surface height at ship's nose	meters
10	Significant Wave Height (Hs)	Statistical average height of waves in the area	meters
11	Peak Wave Period (Tp)	Time interval between consecutive wave crests	seconds
12	Wave Direction	Compass direction waves are coming from	degrees (0°–360°)

Wind Sensor

ROS message: Custom topic or std_msgs/msg/String | **Published frequency:** 10 Hz

#	Parameter	Description	Unit
13	Wind Speed	Over-water wind velocity	m/s
14	Wind Direction	Compass direction wind is coming from	degrees (0°–360°)

Ship Static Parameters

Publish behavior: Published once at startup

#	Parameter	Description	Unit	Source / Calculation
15	Length	Ship length overall / length between perpendiculars (LBP)	meters	Ship principal particulars

16	Beam	Ship maximum width	meters	Ship principal particulars
17	Draft	Ship depth underwater	meters	Ship loading condition / hydrostatic data
18	KG	Center of gravity height above keel	meters	Ship loading condition / stability book
19	GM	Metacentric stability height	meters	Ship loading condition / stability calculation
20	Displacement	Total vessel mass/displacement	tones	Ship load computer / hydrostatic table
21	Block Coefficient (Cb)	Hull fullness coefficient	dimensionless	Ship hydrostatic characteristics
22	Freeboard	Height of the weather deck above the waterline	meters	Distance from waterline to main deck
23	Air Draft	Height from waterline to the ship's highest mast point	meters	Total static height minus draft
24	Number of Propellers	Total number of propulsion shafts/propellers	integer	Vessel configuration

Maneuvering model:

Parameter	Sensor	Description	Data Type
Latitude	GPS / GNSS Receiver	Geographic latitude of the vessel	Float (°)

Longitude	GPS / GNSS Receiver	Geographic longitude of the vessel	Float (°)
Altitude	GPS / GNSS Receiver	Height above mean sea level (optional)	Float (m)
Speed Over Ground (SOG)	GPS / GNSS Receiver	Vessel speed relative to the Earth's surface	Float (knots or m/s)
Course Over Ground (COG)	GPS / GNSS Receiver	Direction of travel over the ground	Float (°)
GPS Timestamp	GPS / GNSS Receiver	UTC timestamp of the GPS reading	DateTime

Parameter	Sensor	Description	Data Type
Heading	Gyro Compass	Vessel heading with respect to True North	Float (°)
Heading Rate	Gyro Compass / INS	Rate of change of heading	Float (°/s)
Roll Angle	INS	Rotation about the longitudinal axis	Float (°)
Pitch Angle	INS	Rotation about the transverse axis	Float (°)
Yaw Rate	INS / IMU	Angular velocity about the vertical axis	Float (rad/s or °/s)

Parameter	Sensor	Description	Data Type
Surge Speed	Doppler Speed Log	Forward speed through water	Float (m/s)
Sway Speed	Doppler Speed Log	Sideways speed through water	Float (m/s)
Speed Log Quality	Doppler Speed Log	Quality indicator of measured speed	Integer

Parameter	Sensor	Description	Data Type
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Rudder Angle	Rudder Feedback Unit	Actual rudder angle	Float (°)
Rudder Command	Steering Gear / Autopilot	Commanded rudder angle	Float (°)
Rudder Rate	Steering Gear	Rate of rudder movement	Float (°/s)

Parameter	Sensor	Description	Data Type
Propeller RPS	Engine Control System	Propeller rotational speed	Float (rps)
Engine RPM	Engine Monitoring System	Engine rotational speed	Float (rpm)
Engine Power	Engine Monitoring System	Engine output power	Float (kW)
Engine Torque	Engine Monitoring System	Shaft torque generated by engine	Float (kNm)
Engine Direction	Engine Control System	Ahead or Astern operation	String

Parameter	Sensor	Description	Data Type
Shaft Power	Shaft Power Meter	Power transmitted through shaft	Float (kW)
Shaft Torque	Shaft Power Meter	Torque transmitted to propeller	Float (kNm)
Shaft RPM	Shaft Power Meter	Rotational speed of shaft	Float (rpm)

Parameter	Sensor	Description	Data Type
Wind Speed	Anemometer	Relative or true wind speed	Float (m/s)
Wind Direction	Wind Vane	Direction from which wind is blowing	Float (°)
Air Temperature	Weather Station	Ambient air temperature	Float (°C)
Atmospheric Pressure	Weather Station	Atmospheric pressure	Float (hPa)

Parameter	Sensor	Description	Data Type
Current Speed	ADCP / Port Monitoring System	Speed of water current	Float (m/s)
Current Direction	ADCP / Port Monitoring System	Direction of water current	Float (°)

Parameter	Sensor	Description	Data Type
Water Depth	Echo Sounder	Total water depth below vessel	Float (m)
Under Keel Clearance	Echo Sounder	Distance between keel and seabed	Float (m)

Parameter	Source	Description	Data Type
Port Bank Distance	ENC / GIS	Distance from vessel to port-side bank	Float (m)
Starboard Bank Distance	ENC / GIS	Distance from vessel to starboard-side bank	Float (m)
Channel Width	ENC / GIS	Width of navigable channel	Float (m)
Fairway Heading	ENC / GIS	Direction of channel centerline	Float (°)

Parameter	Sensor / Source	Description	Data Type
Displacement	Loading Computer	Current vessel displacement	Float (tonnes)
Forward Draft	Draft Sensor	Draft at forward perpendicular	Float (m)
Aft Draft	Draft Sensor	Draft at aft perpendicular	Float (m)
Mean Draft	Loading Computer	Average draft of vessel	Float (m)
Trim	Loading Computer	Difference between aft and forward draft	Float (m)
Heel Angle	Stability System	Inclination of vessel	Float (°)

Parameter	Source	Description	Data Type
Length Between Perpendiculars (LPP)	Ship Particulars / GA Drawing	Reference length used in MMG model	Float (m)
Length Overall (LOA)	Ship Particulars	Total length of vessel	Float (m)
Beam	Ship Particulars	Maximum breadth of vessel	Float (m)
Block Coefficient (Cb)	Hydrostatic Data	Measure of hull fullness	Float
Vessel Mass	Hydrostatic Data	Total vessel mass	Float (kg)
Longitudinal Centre of Gravity (LCG)	Stability Booklet	Longitudinal position of vessel CG	Float (m)

Propeller Diameter	Propeller Specification	Diameter of propeller	Float (m)
Rudder Area	Rudder Drawing	Projected rudder area	Float (m ²)
Rudder Height	Rudder Drawing	Vertical span of rudder	Float (m)
Rudder Chord	Rudder Drawing	Chord length of rudder	Float (m)

Bank Effect

Parameter	Description	Sensors	Unit
Water depth	Total water depth	Echo Sounder	m
sog_knots	Speed Over Ground (used to compute surge velocity u)	GPS Receiver or Speed Log	knots
cog_deg	Course Over Ground	GPS Receiver	degrees (°)
heading_deg	Ship heading	Gyro Compass	degrees (°)
lat	Latitude of vessel	GPS Receiver	decimal degrees
lon	Longitude of vessel	GPS Receiver	decimal degrees
Water Density	Hydrometer	Hydrometer	kg/m ³
Times tamp	Time of the data sample	VDR system clock	float

Drift Model

Parameter	Unit	Description	Source/Sensor
surge_ms	m/s	Forward speed of the vessel	GPS, INS, Simulator
sway_ms	m/s	Sideways (lateral) speed of the vessel	INS, Simulator
yaw_rate_rads	rad/s	Rate at which the vessel is turning	Gyroscope, IMU, Simulator
heading_psi_rad	rad	Current vessel heading angle	Gyrocompass, INS, Simulator
x_earth_m	m	Vessel North/X position	GPS, INS, Simulator
y_earth_m	m	Vessel East/Y position	GPS, INS, Simulator
track_start_x	m	Previous waypoint X coordinate	Navigation System, Route Planner
track_start_y	m	Previous waypoint Y coordinate	Navigation System, Route Planner
track_end_x	m	Next waypoint X coordinate	Navigation System, Route Planner
track_end_y	m	Next waypoint Y coordinate	Navigation System, Route Planner
current_speed_ms	m/s	Speed of river/tidal current	Current Sensor, Environmental Model, Simulator
current_direction_deg	deg	Direction towards which current flows	Current Sensor, Environmental Model, Simulator
wind_speed_ms	m/s	Wind speed	Anemometer, Weather Model, Simulator
wind_direction_deg	deg	Direction from which wind originates	Wind Sensor, Weather Model, Simulator
wave_height_m	m	Significant wave height	Wave Radar, Wave Model, Simulator

wave_period_s	s	Time period between successive waves	Wave Model, Simulator
bank_distance_port_m	m	Distance from vessel to port-side bank	Electronic Chart (ECDIS), Channel Model, Simulator
bank_distance_stbd_m	m	Distance from vessel to starboard-side bank	Electronic Chart (ECDIS), Channel Model, Simulator
dt	s	Time interval between successive updates	Simulator Clock/System Timer